Abhi Chandra Yayi

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EDUCATION

B.S. Information Technology — **August 2022 – May 2026**
Arizona State University (**GPA**: 3.98 | **Dean's List**: 7 semesters) — *Tempe, Arizona*

- Certifications: Back-End Developer Professional Certificate (Meta, Nov 2025); Machine Learning Specialization (DeepLearning.AI, Aug 2025)
- Coursework: Object-Oriented Programming, Database Management, Deep Learning, Agile Development

TECHNICAL SKILLS

Programming Languages: Python, R, SQL, NoSQL, Java, JavaScript, HTML, CSS, Bash, Go
Libraries and Frameworks:Django, Flask, RESTful APIs, React, Scikit-Learn, PyTorch
Tools: Linux, Git, Docker, Kubernetes, Alteryx, Salesforce, Tableau, AWS (ECS, Lambda, SQS), Claude Code
AI/ML: RAG, Vector Search, LoRA/QLoRA Fine-Tuning, Multimodal LLMs, Multi-Agent Tool-Calling, MLOps, MCP

EXPERIENCE

Software Engineer — **July 2026 – Present**
Gabriel AI — *San Francisco, CA*

- Owned the end-to-end redesign and **production** launch of the company's **React/Django** platform, gathering **cross-team** requirements, auditing the architecture, and delivering a **scalable**, production-ready experience.
- Architected and implemented an **asynchronous** backend pipeline using secure REST APIs, **background workers**, queue-based **job orchestration**, and **fault-tolerant** processing for AI voicemail generation.
- Containerized and **deployed** the application with **Docker** and **AWS**, integrating **Firebase** and third-party **APIs** to improve system reliability, maintainability, and production readiness.

Data Science Intern — **June 2025 – July 2025**
Food Forest AI — *Philadelphia, PA*

- Engineered a Python-based data-cleaning **automation** pipeline using **regex** pattern matching, reducing manual processing time by **80%** and improving downstream data accuracy.
- Audited and validated large-scale datasets via **SQL**, identifying duplicates, null values, and schema inconsistencies to improve data ingestion reliability by **40%**.
- Built an **NLP** classification pipeline using embeddings to categorize company descriptions.

PROJECTS

Smart Finance Tracker *(Django, Python, HTML, CSS, MySQL, Git, AWS, OpenAI)*

- Engineered a **full-stack** smart bookkeeping system that enables users to track transactions, categorize spending, and visualize financial insights through interactive dashboards.
- Designed a scalable 3-tier system architecture using **Django/Bootstrap** presentation layer, Django application layer, **MySQL** on **AWS** RDS data layer and deployed via AWS EC2 and VPC.
- Integrated an AI expense-categorization feature using structured **prompt engineering** with an OpenAI **API call**, automatically classifying transactions and reducing manual labeling time by about **65%**.

Agentic Research Assistant *(LangGraph, LangChain, FastAPI, Qdrant, MySQL, Docker, LangSmith)*

- Developed an LLM-powered agentic **RAG** platform ingesting documents to answer complex analytical questions with **cited**, multi-step responses via tool-calling AI **agents**.
- Architected **MCP**-connected **LangGraph** workflows for vector retrieval, SQL querying, Python-based analysis, citation validation, and report generation using **FastAPI, Qdrant, MySQL**, and **Docker**.
- Reduced simulated manual research time by ~**80%** on benchmark tasks, using **LangSmith** for LLM **evaluation** across retrieval quality, faithfulness, latency, and citation accuracy.

AI Token Cost Manager *(Java, Spring Boot, Redis, PostgreSQL, Docker)*

- Currently building a **middleware** gateway that intercepts outbound **LLM API** calls, classifies request complexity, and routes requests to most cost-efficient capable model, eliminating unnecessary use of expensive frontier models.
- Engineered with **Java**, **Spring Boot**, and **Redis** atomic operations for **concurrent** per-user budget enforcement, async usage logging via **background workers**, and **PostgreSQL** for full usage history and audit logging.
- Targeting **60%+** reduction in simulated AI API costs through intelligent **model routing** across GPT-4o, GPT-4o-mini, Claude Sonnet, and Claude Haiku, with real-time cost analytics surfaced via a live dashboard.